PROBLEM NS13

Porous Rotating Cylinders with Radial Flow

Original statement · NS13.pdf

Radial throughflow | Power-law azimuthal velocity | Pressure difference | Temperature profile

OFFICIAL PROBLEM STATEMENT

A liquid of density ρ, viscosity μ, thermal conductivity k, and specific heat c occupies the gap formed between two porous coaxial cylinders of radius R_i and R_o that are rotating with angular velocities Ω_i and Ω_o, respectively. The liquid is injected through the surface of the inner cylinder with normal velocity component $v_r = V_i$ and is removed through the outer surface. The temperatures on the inner and outer surfaces are T_i and T_o, respectively. The resulting steady, planar, axially symmetric flow field is described in terms of $v_r(r)$, $v_\theta(r)$, $p(r)$, and $T(r)$.

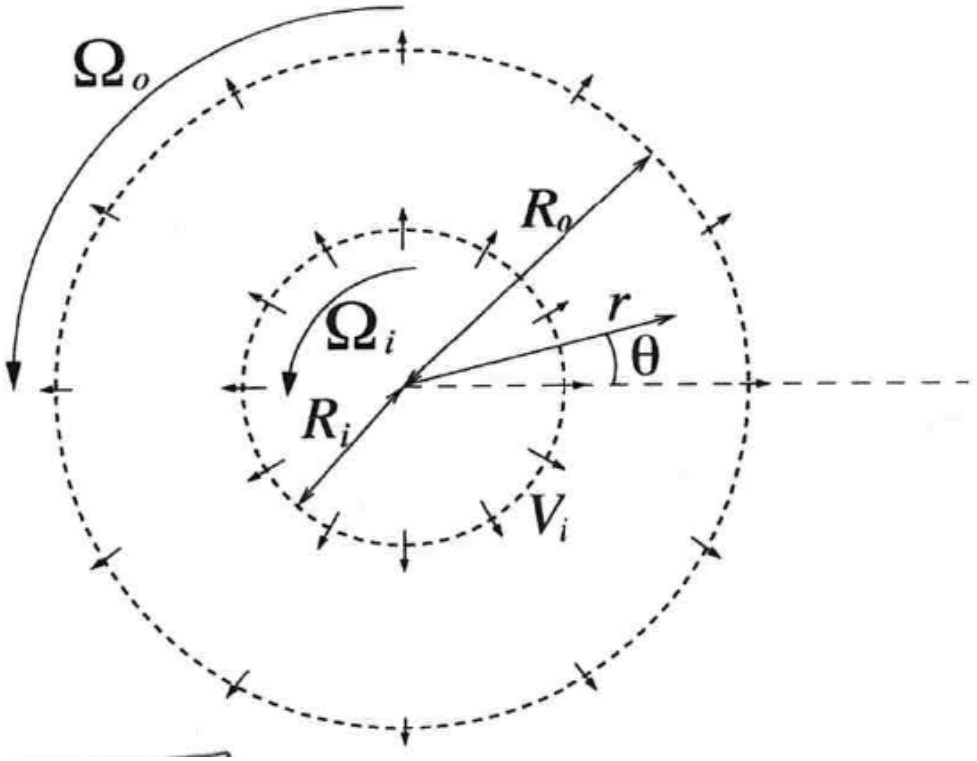
1. Determine the radial velocity component $v_r(r)$. Obtain in particular its value on the surface of the outer cylinder $v_r(R_o)$.

2. Obtain the azimuthal velocity component $v_\theta(r)$. To that end, write the θ-component of the momentum equation in terms of the parameter $\mathrm{Re} = \rho V_i R_i/\mu$ and look for solutions of the form $v_\theta \propto r^\alpha$, demonstrating that the general solution can be expressed as $v_\theta = C_1/r + C_2\, r^{\mathrm{Re}+1}$. Obtain the values of the integration constants C_1 and C_2.

3. Calculate $p(r) - p_i$, the difference of the pressure $p(r)$ from its inner-surface value $p_i = p(R_i)$.

4. Write the equation that determines the temperature distribution $T(r)$, including the appropriate boundary conditions. Solve the problem assuming that the effect of viscous dissipation is negligibly small.

The sheet is headed "INTRODUCTION TO FLUID MECHANICS, MIDTERM (Nov 11, 2013)".



Statement figure (crop of NS13.pdf, p. 1): porous cylinders of radii R_i and R_o rotating with Ω_i and Ω_o, injection velocity V_i through the inner surface, polar coordinates r, θ

`Source: NS13.pdf (page 1). Notes.pdf, Chapter 4: continuity Eq. 4.40, momentum Eq. 4.41 and energy Eq. 4.42 in cylindrical coordinates, viscous dissipation $\phi_v$ in cylindrical coordinates.`

Worked solution (4-phase method)

`PHASE 1`

Hypotheses, Data and Unknowns

- Steady, planar ($\partial/\partial z = 0, v_z = 0$), axisymmetric ($\partial/\partial\theta = 0$) flow of a perfect liquid (constant ρ, μ, k, c) between two porous cylinders: $\bar{v} = v_r(r)\,\bar{e}_r + v_\theta(r)\,\bar{e}_\theta, p = p(r)$, $T = T(r)$.
- No mass force enters the plane of motion; no chemical reaction and no radiation.
- The inner cylinder ($r = R_i$) injects the liquid with $v_r = V_i > 0$ and the outer one removes it; both rotate rigidly with their angular velocities (no-slip).
- Data: $[\rho,\ \mu,\ k,\ c,\ R_i,\ R_o,\ \Omega_i,\ \Omega_o,\ V_i,\ T_i,\ T_o,\ p_i]$. Unknowns: $v_r, v_\theta, p - p_i, T$.

- Dimensionless numbers: $\mathrm{Re} = \rho V_i R_i / \mu$ and $\mathrm{Pr} = \mu c / k$.

Symbol	Meaning	SI unit
R_i, R_o	inner and outer radii	m
Ω_i, Ω_o	angular velocities of the cylinders	rad/s
V_i	injection velocity at the inner surface	m/s
T_i, T_o	surface temperatures	K
Re, Pr	injection Reynolds number, Prandtl number	1
C_1, C_2	integration constants of v_θ	m^2/s, $1/\mathrm{s} \cdot \mathrm{m}^{-\mathrm{Re}}$

PHASE 2

Frames, Field Form and Boundary Conditions

- Cylindrical frame (r, θ, z) centred on the common axis, with the orthonormal right-handed basis $\{\bar{e}_r, \bar{e}_\theta, \bar{e}_z\}$ fixed to the laboratory (inertial).
- Field form: $\bar{v} = v_r(r)\,\bar{e}_r + v_\theta(r)\,\bar{e}_\theta, p = p(r), T = T(r)$.

Location	Condition	Origin
$r = R_i$	$v_r = V_i, v_\theta = \Omega_i R_i$	injection through the inner surface, no-slip
$r = R_o$	$v_\theta = \Omega_o R_o$	no-slip on the outer cylinder
$r = R_i$	$p = p_i, T = T_i$	reference pressure, inner temperature
$r = R_o$	$T = T_o$	outer temperature

PHASE 3

Step-by-Step Derivation

3.1 Continuity (item 1)

$$\frac{1}{r}\frac{d(r v_r)}{dr} + \underbrace{\frac{1}{r}\frac{\partial v_\theta}{\partial \theta}}_{\partial_\theta = 0} + \underbrace{\frac{\partial v_z}{\partial z}}_{v_z = 0} = 0 \implies r\,v_r = \mathrm{const} = R_i V_i \qquad (1)$$

where the constant is fixed with $v_r(R_i) = V_i$:

$$\boxed{v_r(r) = \frac{R_i V_i}{r}, \qquad v_r(R_o) = \frac{R_i V_i}{R_o}} \tag{2}$$

3.2 Azimuthal momentum (item 2)

The θ component of (4.41) with $\partial_t = \partial_\theta = \partial_z = 0$, no mass force and $\partial p/\partial\theta = 0$:

$$\rho\left(v_r \frac{dv_\theta}{dr} + \frac{v_r v_\theta}{r}\right) = \mu \frac{d}{dr}\left[\frac{1}{r}\frac{d(r v_\theta)}{dr}\right] \tag{3}$$

The left side is $\rho v_r \, \dfrac{1}{r}\dfrac{d(r v_\theta)}{dr}$, since $dv_\theta/dr + v_\theta/r = r^{-1} d(r v_\theta)/dr$. With $v_r = R_i V_i / r$ and $\mathrm{Re} = \rho V_i R_i / \mu$:

$$\frac{d}{dr}\left[\frac{1}{r}\frac{d(r v_\theta)}{dr}\right] - \mathrm{Re}\,\frac{1}{r}\left(\frac{dv_\theta}{dr} + \frac{v_\theta}{r}\right) = 0 \tag{4}$$

Try $v_\theta = r^\alpha$: $d(r v_\theta)/dr = (\alpha+1) r^\alpha$, so $r^{-1} d(r v_\theta)/dr = (\alpha+1) r^{\alpha-1}$ and its derivative is $(\alpha+1)(\alpha-1) r^{\alpha-2}$; the second term of (4) is $\mathrm{Re}\,(\alpha+1) r^{\alpha-2}$. Substituting and dividing by $r^{\alpha-2}$:

$$(\alpha+1)(\alpha - 1 - \mathrm{Re}) = 0 \;\Longrightarrow\; \alpha = -1 \quad \text{or} \quad \alpha = \mathrm{Re} + 1 \tag{5}$$

The general solution is therefore

$$v_\theta = \frac{C_1}{r} + C_2\, r^{\mathrm{Re}+1} \tag{6}$$

The boundary conditions $v_\theta(R_i) = \Omega_i R_i$ and $v_\theta(R_o) = \Omega_o R_o$, each multiplied by R_i and R_o respectively, give

$$C_1 + C_2 R_i^{\mathrm{Re}+2} = \Omega_i R_i^2, \qquad C_1 + C_2 R_o^{\mathrm{Re}+2} = \Omega_o R_o^2 \tag{7}$$

Subtracting, $C_2\left(R_o^{\mathrm{Re}+2} - R_i^{\mathrm{Re}+2}\right) = \Omega_o R_o^2 - \Omega_i R_i^2$, and then $C_1 = \Omega_i R_i^2 - C_2 R_i^{\mathrm{Re}+2}$:

$$\boxed{C_1 = \frac{R_i^2 R_o^2\left(\Omega_i R_o^{\mathrm{Re}} - \Omega_o R_i^{\mathrm{Re}}\right)}{R_o^{\mathrm{Re}+2} - R_i^{\mathrm{Re}+2}}, \qquad C_2 = \frac{\Omega_o R_o^2 - \Omega_i R_i^2}{R_o^{\mathrm{Re}+2} - R_i^{\mathrm{Re}+2}}} \tag{8}$$

3.3 Pressure difference (item 3)

The r component of (4.41): the viscous term $\mu\, d[r^{-1} d(r v_r)/dr]/dr$ is zero by (1), so

$$\rho\left(v_r \frac{dv_r}{dr} - \frac{v_\theta^2}{r}\right) = -\frac{dp}{dr} \implies \frac{dp}{dr} = \rho\left[\frac{v_\theta^2}{r} + \frac{(R_i V_i)^2}{r^3}\right] \tag{9}$$

since $v_r\, dv_r/dr = (R_i V_i/r)(-R_i V_i/r^2) = -(R_i V_i)^2/r^3$. Squaring (6), $v_\theta^2/r = C_1^2 r^{-3} + 2C_1 C_2 r^{\mathrm{Re}-1} + C_2^2 r^{2\mathrm{Re}+1}$. Integrating from R_i to r (antiderivatives $-C_1^2/(2r^2)$, $2C_1C_2 r^{\mathrm{Re}}/\mathrm{Re}$, $C_2^2 r^{2\mathrm{Re}+2}/(2\mathrm{Re}+2)$ and $-(R_iV_i)^2/(2r^2)$) and using Barrow's rule:

$$\boxed{\frac{p(r) - p_i}{\rho} = \frac{C_1^2 + (R_i V_i)^2}{2}\left(\frac{1}{R_i^2} - \frac{1}{r^2}\right) + \frac{2C_1 C_2}{\mathrm{Re}}\left(r^{\mathrm{Re}} - R_i^{\mathrm{Re}}\right) + \frac{C_2^2}{2(\mathrm{Re}+1)}\left(r^{2\mathrm{Re}+2} - R_i^{2\mathrm{Re}+2}\right)} \tag{10}$$

(for $\mathrm{Re} \neq 0$; for $\mathrm{Re} = 0$ the second term is $2C_1C_2\ln(r/R_i)$).

3.4 Temperature (item 4)

The energy equation (4.42) with $v_r = R_i V_i/r$ and the dissipation ϕ_v of the cylindrical expression (Notes.pdf) with $v_z = 0$, and the boundary conditions:

$$\rho c\, v_r \frac{dT}{dr} = \phi_v + \frac{k}{r}\frac{d}{dr}\left(r\frac{dT}{dr}\right), \quad \phi_v = \mu\left\{2\left[\left(\frac{dv_r}{dr}\right)^2 + \left(\frac{v_r}{r}\right)^2\right] + \left[r\frac{d}{dr}\left(\frac{v_\theta}{r}\right)\right]^2\right\}, \quad T(R_i) = T_i,\ T(R_o) = \tag{11}$$

Neglecting ϕ_v and multiplying by r/k, with $\rho c R_i V_i/k = \mathrm{Pr}\,\mathrm{Re}$:

$$\frac{d}{dr}\left(r\frac{dT}{dr}\right) - \mathrm{Pr}\,\mathrm{Re}\,\frac{dT}{dr} = 0 \tag{12}$$

Try $T = r^{\beta}$: $\beta^2 - \mathrm{Pr}\,\mathrm{Re}\,\beta = 0$, so $\beta = 0$ or $\beta = \mathrm{Pr}\,\mathrm{Re}$ and $T = B_1 + B_2\, r^{\mathrm{Pr}\,\mathrm{Re}}$. The conditions $T(R_i) = T_i$ and $T(R_o) = T_o$ give $B_2\left(R_o^{\mathrm{Pr}\,\mathrm{Re}} - R_i^{\mathrm{Pr}\,\mathrm{Re}}\right) = T_o - T_i$:

$$\boxed{T(r) = T_i + (T_o - T_i)\,\frac{r^{\mathrm{Pr}\,\mathrm{Re}} - R_i^{\mathrm{Pr}\,\mathrm{Re}}}{R_o^{\mathrm{Pr}\,\mathrm{Re}} - R_i^{\mathrm{Pr}\,\mathrm{Re}}}} \tag{13}$$

PHASE 4

Interpretation, Limits and Dimensional Check

- **Physical meaning.** The radial throughflow convects angular momentum and heat outwards: C_1/r is a free vortex (rv_θ constant) and $C_2\, r^{\mathrm{Re}+1}$ is the term that the radial convection modifies: it reduces to the rigid rotation $C_2\, r$ for $\mathrm{Re} = 0$ and grows faster with r for $\mathrm{Re} > 0$. For $\mathrm{Re} \to 0$ (no injection) the solution reduces to the Couette profile between rotating cylinders, $v_\theta = C_1/r + C_2\, r$ with $C_1 = R_i^2 R_o^2(\Omega_i - \Omega_o)/(R_o^2 - R_i^2)$ and $C_2 = (\Omega_o R_o^2 - \Omega_i R_i^2)/(R_o^2 - R_i^2)$, as in problem NS2 (with the roles of the two constants exchanged).
- **Limits.** For $\Omega_i = \Omega_o = 0$: $C_1 = C_2 = 0$, $v_\theta = 0$ and the pressure (10) is $\rho(R_i V_i)^2(1/R_i^2 - 1/r^2)/2$, the Bernoulli pressure recovery of the radial flow, $p - p_i = \frac{\rho}{2}\left(V_i^2 - v_r^2\right) > 0$: the pressure rises outwards as the liquid slows down. For $\mathrm{Pr}\,\mathrm{Re} \to 0$, $r^{\mathrm{Pr}\,\mathrm{Re}} \simeq 1 + \mathrm{Pr}\,\mathrm{Re}\,\ln r$ and (13) tends to the pure conduction profile $T = T_i + (T_o - T_i)\ln(r/R_i)/\ln(R_o/R_i)$.
- **Dimensional check.** $R_i V_i/r$ is a velocity. C_1 has units m^2/s and $C_2\, r^{\mathrm{Re}+1}$ has units m/s. In (10) every term has units of $\rho\,(\mathrm{m/s})^2$, that is Pa. $\mathrm{Pr}\,\mathrm{Re}$ is dimensionless, so the argument of (13) is dimensionless.
- **Boundary check.** From (6)-(8), $v_\theta(R_i) = \Omega_i R_i$ and $v_\theta(R_o) = \Omega_o R_o$; from (13), $T(R_i) = T_i$ and $T(R_o) = T_o$.